

LABUAN, Malaysia

WMO: 964650

Lat: 5.301N

Lon: 115.250E

Elev: 31

StdP: 100.96

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.4	23.9	21.0	15.7	28.9	21.8	16.5	28.1	7.8	28.1	7.3	28.1	1.8	240	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.8	33.1	26.5	32.6	26.5	32.1	26.5	28.0	31.1	27.7	30.7	27.3	30.4	2.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.1	23.0	30.2	26.9	22.6	29.9	26.3	21.9	29.3	90.2	31.4	88.4	30.9	87.0	31.0	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.7	5.0	21.8	34.7	1.3	1.3	20.8	35.6	20.1	36.4	19.3	37.2	18.4	38.1		
(5)			21.0	28.8	1.3	0.8	20.1	29.4	19.3	29.8	18.6	30.3	17.6	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	27.6	27.6	28.2	28.7	28.8	28.4	28.2	28.3	28.2	27.9	27.9	27.8
	DBStd	1.09	0.99	1.00	0.93	0.96	1.06	1.05	1.12	1.16	1.03	1.06	0.90	1.03
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6619	544	493	563	560	582	553	566	568	545	555	538	553
	CDD18.3	3577	286	259	305	310	323	303	307	309	295	297	288	295
	CDH23.3	39546	2941	2614	3286	3609	3842	3466	3441	3518	3300	3221	3129	3181
CDH26.7	13266	784	676	1041	1358	1503	1259	1208	1274	1133	1038	1002	990	
Wind	WSAvg	2.0	2.7	2.8	2.3	1.8	1.6	1.6	1.6	1.7	1.7	1.9	1.8	2.1
Precipitation	PrecAvg	3282	249	128	157	234	292	288	274	283	325	384	367	302
	PrecMax	4656	747	384	404	479	514	572	608	606	880	682	619	504
	PrecMin	2308	23	25	41	100	143	131	95	96	130	169	210	77
	PrecStd	542	179	101	90	99	96	123	118	140	136	149	107	125
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	32.4	33.1	33.8	33.8	33.1	33.1	33.2	33.0	32.8	32.5	32.2
		MCWB	26.1	26.2	26.2	26.7	27.0	26.6	26.2	26.1	26.2	26.5	26.7	26.5
	2%	DB	31.0	31.2	32.0	32.5	32.8	32.2	32.2	32.2	32.1	31.9	31.7	31.6
		MCWB	26.2	26.2	26.2	26.7	26.9	26.6	26.3	26.2	26.2	26.4	26.6	26.5
	5%	DB	30.2	30.4	31.2	31.9	32.1	31.9	31.7	31.8	31.4	31.1	31.1	31.0
		MCWB	26.0	26.0	26.2	26.6	26.7	26.6	26.3	26.2	26.2	26.4	26.5	26.4
	10%	DB	29.8	29.8	30.6	31.1	31.5	31.1	31.0	31.1	30.9	30.6	30.4	30.2
		MCWB	25.9	25.8	26.0	26.6	26.7	26.5	26.3	26.3	26.3	26.2	26.3	26.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.2	27.4	27.6	28.1	28.3	28.2	27.9	28.0	27.9	27.8	28.1	27.7
		MCDB	29.9	30.4	31.1	31.2	31.5	31.2	30.9	30.9	30.9	30.8	30.8	30.3
	2%	WB	26.9	26.8	27.1	27.6	27.8	27.7	27.3	27.4	27.2	27.1	27.2	27.1
		MCDB	29.5	29.7	30.2	30.7	31.0	30.8	30.3	30.4	30.3	30.3	30.1	29.9
	5%	WB	26.5	26.4	26.7	27.2	27.5	27.2	26.9	27.0	26.8	26.7	26.8	26.7
		MCDB	29.0	29.1	29.6	30.3	30.6	30.3	30.0	30.1	29.9	29.8	29.7	29.5
	10%	WB	26.2	26.1	26.4	26.9	27.1	26.8	26.6	26.6	26.5	26.4	26.5	26.4
		MCDB	28.6	28.6	29.1	29.9	30.2	29.8	29.5	29.7	29.5	29.3	29.3	29.1
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.0	5.6	5.9	5.8	5.9	5.8	5.9	5.9	5.6	5.6	5.3
		MCDBR	5.3	5.7	6.1	6.4	6.3	6.3	6.2	6.3	6.3	6.2	6.1	5.9
		MCWBR	2.4	2.3	2.3	2.5	2.5	2.7	2.5	2.6	2.7	2.6	2.6	2.5
	5% WB	MCDBR	5.1	5.4	5.8	6.1	6.2	6.2	6.0	6.1	6.1	6.0	5.9	5.6
		MCWBR	2.5	2.4	2.5	2.6	2.7	2.9	2.8	2.9	2.9	2.8	2.7	2.6
Clear-Sky Solar Irradiance	taub	0.410	0.418	0.422	0.424	0.408	0.411	0.409	0.444	0.444	0.453	0.413	0.409	
	taud	2.518	2.468	2.462	2.449	2.510	2.515	2.514	2.388	2.383	2.368	2.502	2.524	
	Ebn at Noon	910	909	903	884	880	866	871	856	869	866	900	905	
	Edh at Noon	108	116	117	117	107	104	105	122	125	126	109	106	
All-Sky Solar Radiation	RadAvg	4.87	5.53	5.89	5.99	5.50	5.16	5.15	5.33	5.20	4.89	4.93	4.75	
	RadStd	0.44	0.45	0.35	0.26	0.25	0.24	0.24	0.24	0.33	0.35	0.32	0.39	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-0.11	N/A	+0.23	+0.34	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.15	N/A	N/A	+0.29	+0.26	+0.23	N/A	N/A	N/A	N/A

Nomenclature: See separate page